Fiscal Year:	FY 2016	**Task Last Updated:**	FY 05/02/2016
PI Name:	Limoli, Charles Ph.D.		
Project Title:	NSCOR: Mechanisms Underlying Charged Particle-Induced Disruption of CNS Function		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	92697-2695	**Congressional District:**	45
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2013-14 HERO NNJ13ZSA002N-NSCOR Radiation
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No. of PhD Candidates:	2	**No. of Master' Degrees:**	
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No. of Bachelor's Candidates:	4	**Monitoring Center:**	NASA JSC
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Flight Program:			
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Task Description:	Humans exposed to the space radiation environment encounter a different type of radiation than that found on Earth. Highly energetic particles can traverse the body, including the brain, and leave a path of damage that is difficult to repair and disrupts function. One consequence of this exposure is the development of radiation-induced cognitive dysfunction. The focus of this application is to resolve the mechanisms leading to radiation-induced cognitive dysfunction, in order to improve risk estimates for early and late CNS (central nervous system) effects caused by exposure to the space radiation environment. We have found that radiation impacts excitatory and inhibitory circuitry in the brain differently, and leads to significant disruptions to the structural integrity of neurons while altering the levels and activity of the proteins that regulate neuronal function. The situation is further complicated by the fact that different regions of the brain exhibit differential sensitivity to charged particle irradiation. Based on the foregoing we propose a comprehensive series of mechanistic studies analyzing the biochemical, structural, and electrophysiologic changes induced by radiation exposure that alter function to impact cognition. Carefully selected cognitive tasks and a wide variety of approaches will be used to critically analyze microcircuits in the brain and how charged particle-induced changes are transmitted via unique signaling pathways between cells that can alter gene expression profiles within cells. Collectively these studies will span a radiation effects pathway from synapse to cognition, with an overarching goal of determining whether charged particle irradiation leads to alterations in the excitatory and inhibitory activity of the brain at a global and/or region specific level. Our studies will therefore elucidate the functional consequences of radiation-induced changes on cognition that impose risk on mission critical activities and long-term cognitive health.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Work will characterize several of the adverse effects of ionizing radiation on the CNS and help to define the early and late CNS risks associated with exposure to the space radiation environment. There is a real possibility that many of the consequences of cosmic radiation exposure will extend over many years and impact (possibly negatively) neurocognitive health after return to Earth. Thus, research elucidating the mechanistic basis of charged particle-induced effects in the irradiated brain may well have functional importance to terrestrial situations where the public may be exposed to ionizing radiation from a variety of potential sources.
Task Progress:	Scientific Overview The CNS NASA Specialized Center of Research (NSCOR) is organized centrally at the University of California, Irvine (UCI: PI & Director, Limoli) and is conducted between 3 other universities, including Stanford (PI and Co-director, Soltesz), Loma Linda University (LLU: PIs Nelson & Vlkolinsky), and Eastern Virginia Medical School (EVMS: PI,Britten). All proposed work is focused under an overarching hypothesis that seeks to define the mechanisms contributing to charged particle–induced disruptions to CNS functionality that impact risk over acute (mission critical) and more chronic (terrestrial) timeframes. The unique composition of the NSCOR team provides the unprecedented opportunity to map a radiation effects pathway from synapse to cognition. These efforts will characterize the changes in synaptic integrity, structural parameters of neurons, electrophysiologic properties of connectivity and neurotransmission in defined circuits, and a variety of cognitive tasks that interrogate multiple cortical and hippocampal regions of the brain at various times after charged particle exposure. These studies will also be coordinated with larger field recordings of neuronal ensembles within distinct cortical regions, and with studies focused on the role of activated microglia and secreted exosomes in regulating the radiation response of the brain and how epigenetics controls the expression of key genes known to impact learning and memory. Many of the foregoing studies will be conducted in both mice and rats and all experiments will utilize males and female animals exposed at astronaut relevant ages (6 months for mice, 9 months for rats). Progress to date at the University of California, Irvine: Findings uncovered at UCI have been able to demonstrate direct links between charged particle induced disruptions to cognition and persistent damage to specific neuronal subsets interrogated by our behavioral tasks. Since our seminal paper published in Science Advances last year entitled, “What happens to your brain on the way to Mars,” we have demonstrated that these effects are even more persistent than previously thought. Significant effort has now found that rodents exposed to energetic charged particles exhibit persistent hippocampal and cortical based performance decrements using six independent behavioral tasks administered 12 and 24 weeks after irradiation. Radiation-induced impairments in spatial, episodic, and recognition memory were temporally coincident with deficits in executive function and reduced rates of fear extinction and elevated anxiety. Irradiation caused significant reductions in dendritic complexity, spine density, and altered spine morphology along medial prefrontal cortical neurons known to mediate neurotransmission interrogated by our behavioral tasks. Exposure to charged particles also disrupted synaptic integrity and increased neuroinflammation that persisted more than 6 months after exposure. These new data corroborate our prior work, and provide convincing follow up data that demonstrates why cosmic radiation encountered during deep space travel poses such a real and unique threat to the integrity of neural circuits in the brain. Progress to date at the Stanford University: Within the reporting period under this grant, work in the Soltesz lab at Stanford focused on the centrally important question of how space-relevant irradiation affects neurotransmission in the central nervous system. Recent evidence has confirmed that space relevant doses of protons can elicit persistent disruptions in cognition that can be linked to changes in the structural and synaptic plasticity of mature neurons. However, relatively little is known regarding the impact of proton exposures on circuit specific synaptic transmission and connectivity in the brain. Results demonstrate, for the first time, that low-dose proton irradiation selectively compromises hippocampal GABAergic inhibition, as well as local excitatory connectivity, suggesting that cell type-specific disruptions in hippocampal circuits may mechanistically contribute to radiation-induced cognitive deficits. It is noteworthy that these cell type-specific alterations both indicate an increase inhibitory drive on CA1 PCs that form the major information-processing output channels from the hippocampal formation. Progress to date at the Eastern Virginia Medical School: The initial focus of activities at this subset are to conduct complex cognitive behavioral tasks that interrogate many aspects of executive function (i.e., Attentional Set Shifting, ATSET). To date, 18 irradiated rats (6 rats/dose) and 6 sham-irradiated rats were screened for ATSET performance during February and March 2016. Overall, there were no significant differences in the ATSET performance of the irradiated rats compared to the sham-irradiated rats. However, rats that were exposed to 1.5 cGy 400 MeV/n oxygen ions took almost double the number of attempts to pass compound

	discrimination (CD) (P=0.044, Welch's t-test). These rats have now been sent to Loma Linda University for follow up electrophysiologcic assessments to uncover potential mechanisms accounting for radiation-induced deficits in CD. Progress to date at Loma Linda University: Infrastructure upgrades and preliminary experiments have established our ability to interrogate neocortical regions of rat brain with electrophysiological methods. We have established baseline responses to protons and oxygen ions and have observed changes in synaptic plasticity in the form of altered long-term depression. Interestingly, in rats trained for ATSET the behavioral and electrophysiological decrements in LTD were observed in same radiation groups (0.015 Gy), but not in other groups, suggesting a correlation. Trends have been observed in AMPA-receptor mediated miniature currents in single neurons as well as changes in membrane input resistance that resembles previous results from mouse hippocampus. We are now proceeding to acquire mature male rats for a thorough characterization of proton effects in the medial prefrontal cortex and hipocampus and will investigate the properties of the perirhinal cortex as well.
Bibliography Type:	Description: (Last Updated: 01/14/2026)
Articles in Peer-reviewed Journals	Parihar VK, Allen B, Tran KK, Macaraeg TG, Chu EM, Kwok SF, Chmielewski NN, Craver BM, Baulch JE, Acharya MM, Cucinotta FA, Limoli CL. "What happens to your brain on the way to Mars." Science Advances. 2015 May 1;1(4). http://dx.doi.org/10.1126/sciadv.1400256 ; PubMed PMID: 26180843; PubMed Central PMCID: PMC4500198 , May-2015
Articles in Peer-reviewed Journals	Limoli CL. "Your brain on Mars. Comment." Radiat Res. 2015 Jul;184(1):1-2. Epub 2015 Jun 11. Comment. PubMed PMID: 26066080 ; http://dx.doi.org/10.1667/RR14143.1 , Jul-2015
Articles in Peer-reviewed Journals	Lee SH, Ledri M, Tóth B, Marchionni I, Henstridge CM, Dudok B, Kenesei K, Barna L, Szabó SI, Renkecz T, Oberoi M, Watanabe M, Limoli CL, Horvai G, Soltesz I, Katona I. "Multiple forms of endocannabinoid and endovanilloid signaling regulate the tonic control of GABA release." J Neurosci. 2015 Jul 8;35(27):10039-57. http://dx.doi.org/10.1523/JNEUROSCI.4112-14.2015 ; PubMed PMID: 26157003; PubMed Central PMCID: PMC4495235 , Jul-2015
Articles in Peer-reviewed Journals	Chmielewski NN, Caressi C, Giedzinski E, Parihar VK, Limoli CL. "Contrasting the effects of proton irradiation on dendritic complexity of subiculum neurons in wild type and MCAT mice." Environ Mol Mutagen. 2016 Mar 20. [Epub ahead of print] http://dx.doi.org/10.1002/em.22006 ; PubMed PMID: 26996825 , Mar-2016